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United States Patent	12400886
Kind Code	B2
Date of Patent	August 26, 2025
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Apparatus for transporting substrate and system for treating substrate with the apparatus

Abstract

Provided are a substrate transporting apparatus capable of preventing an increase in temperature of a transporting robot by installing a cooling plate around the transporting robot, and a substrate treating system including the same. The substrate transporting apparatus includes a transporting unit for transporting a substrate; and a cooling plate for controlling a temperature of the transporting unit, wherein the cooling plate is spaced apart from a side surface of the transporting unit and installed as a side wall, or is installed in close contact with the side surface of the transporting unit.

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Appl. No.: 18/088986

Filed: December 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230130873 A1	Apr. 27, 2023

Foreign Application Priority Data

KR	10-2020-0005548	Jan. 15, 2020
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Related U.S. Application Data

Publication Classification

Int. Cl.: **B25J19/00** (20060101); **B65G47/90** (20060101); **B65G69/20** (20060101); **H01L21/67** (20060101); **H01L21/677** (20060101)

U.S. Cl.:

CPC **H01L21/67196** (20130101); **B65G47/90** (20130101); **B65G69/20** (20130101); **H01L21/67748** (20130101); B25J19/0054 (20130101)

Field of Classification Search

CPC: H01L (21/67196); H01L (21/67748); B65G (47/90); B65G (69/20); B25J (19/0054)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a U.S. Continuation of U.S. patent application Ser. No. 17/148,914, filed Jan. 14, 2021, which claims the benefit of Korean Patent Application No. 10-2020-0005548, filed on Jan. 15, 2020, which are hereby incorporated by reference in their entirety into this application.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a substrate transporting apparatus for transporting a substrate and a substrate treating system including the same. More specifically, it relates to a substrate transporting apparatus arranged between a plurality of process chambers and a substrate treating system including the same.

Description of the Related Art

(2) A semiconductor device can be manufactured by forming a predetermined pattern on a substrate. When forming a predetermined pattern on a substrate, a number of processes such as a deposition process, a photo-lithography process, an etching process and a cleaning process may be continuously performed within a FAB.

(3) A number of processes performed when manufacturing a semiconductor device may be done in individual chambers. In this case, the substrate may be transported to each chamber through a transporting robot.

SUMMARY OF THE INVENTION

(4) The transporting robot may use a LM (Linear Motion) guide system when transporting a substrate (e.g., wafer). However, while the transporting robot moves on the LM rail to transport the substrate to the process chamber, heat generated by the driving of the motor may increase the temperature of the transporting robot.

(5) Further, the transporting robot may be arranged between a plurality of process chambers to transport the substrate to each process chamber. Since the process chamber is in a high temperature state to perform a predetermined process, the temperature of the transporting robot may increase due to the process chamber.

(6) When the surface temperature of the transporting robot is increased as described above, it may affect the driving performance of the transporting robot, such as abnormal motions of the transporting robot during the operation, and the lifespan of the transporting robot may be shortened.

(7) The problem to be solved in the present invention is to provide a substrate transporting apparatus capable of preventing an increase in temperature of the transporting robot by installing a cooling plate around the transporting robot, and a substrate treating system including the same.

(8) The problems of the present invention are not limited to the problems mentioned above, and other problems that are not mentioned will be clearly understood by those skilled in the art from the following description.

(9) One aspect of the substrate transporting apparatus of the present invention for achieving the above object comprises a transporting unit for transporting a substrate; and a cooling plate for controlling a temperature of the transporting unit, wherein the cooling plate is spaced apart from a side surface of the transporting unit and installed as a side wall, or installed in close contact with a side surface of the transporting unit.

(10) A flow path, through which coolant flows, may be installed inside the cooling plate.

(11) The cooling plate may comprise a first plate member; a second plate member coupled to the first plate member; and a flow path formed in a space between the first plate member and the second plate member, through which coolant flows.

(12) The flow path may be provided as a pipe inserted into a space between the first plate member and the second plate member, or a groove formed on at least one surface of at least one member of the first plate member and the second plate member.

(13) When the groove is formed in the first plate member, the groove may be formed on a surface of the first plate member that contacts the second plate member when the first plate member and the second plate member are coupled.

(14) When the groove is formed in the second plate member, the groove may be formed on a surface of the second plate member that contacts the first plate member when the first plate member and the second plate member are coupled.

(15) At least one of a thermal compound, a thermal pad, and a thermal grease may be installed on the flow path.

(16) The flow path may be formed in a zigzag shape.

(17) The first plate member and/or the second plate member may be manufactured of a material emitting cool air to outside as a material.

(18) The cooling plate may further comprise a pump for supplying coolant to the flow path; and a valve for controlling a flow of coolant by opening and closing the flow path.

(19) The cooling plate may be further installed in a lower portion of the transporting unit.

(20) When the cooling plate is installed in close contact with a side surface of the transporting unit, the cooling plate is installed using a combination of a first protrusion formed on the cooling plate

and a second protrusion formed on the transporting unit, and the first protrusion and the second protrusion may have different heights.

(21) When the cooling plate is installed in close contact with a side surface of the transporting unit, the cooling plate is installed using a combination of a protrusion formed on one of the cooling plate and the transporting unit, and a groove formed on the other, and the protrusion and the groove may have the same height.

(22) The coolant may be PCW (Process Cooling Water).

(23) The apparatus for transporting a substrate may further comprise a guide rail for providing a moving path of the transporting unit; and a support block coupled to an upper portion of the guide rail to support the transporting unit, and sliding along the guide rail.

(24) Another aspect of the substrate transporting apparatus of the present invention for achieving the above object comprises a transporting unit for transporting a substrate; and a cooling plate for controlling a temperature of the transporting unit, wherein the cooling plate is spaced apart from a side surface of the transporting unit and installed as a side wall, or is installed in close contact with a side surface of the transporting unit, wherein a flow path, through which coolant flows, is installed inside the cooling plate, wherein at least one of a thermal compound, a thermal pad, and a thermal grease is installed on the flow path.

(25) One aspect of the substrate treating system of the present invention for achieving the above object comprises a first process chamber for treating a substrate; a second process chamber for treating the substrate; and a substrate transporting device including a transporting unit for transporting the substrate between the first process chamber and the second process chamber, and a cooling plate for controlling a temperature of the transporting unit, wherein the cooling plate is spaced apart from a side surface of the transporting unit and installed as a side wall, or is installed in close contact with a side surface of the transporting unit.

(26) Details of other embodiments are included in the detailed description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view schematically showing an internal structure of a substrate treating system including a substrate transporting device.

(2) FIG. 2 is a perspective view showing a detailed structure of the substrate transporting device shown in FIG. 1.

(3) FIG. 3 is a plan view schematically showing a substrate transporting device including a cooling plate according to an embodiment of the present invention.

(4) FIG. 4 is a cross-sectional view schematically showing a substrate transporting device including a cooling plate according to an embodiment of the present invention.

(5) FIG. 5 is a perspective view showing a detailed structure of a cooling plate constituting a substrate transporting device according to an embodiment of the present invention.

(6) FIG. 6 is a first exemplary view showing various installation types of pipe members constituting the cooling plate shown in FIG. 5.

(7) FIG. 7 is a second exemplary view showing various installation types of pipe members constituting the cooling plate shown in FIG. 5.

(8) FIG. 8 is a reference diagram for describing the performance of a substrate transporting device including a cooling plate according to an embodiment of the present invention.

(9) FIG. 9 is a cross-sectional view schematically showing a substrate transporting devices including a cooling plate according to another embodiment of the present invention.

(10) FIG. 10 is a first exemplary view for describing various types of coupling between a transporting unit and a cooling plate constituting a substrate transporting device according to an

embodiment of the present invention.

(11) FIG. 11 is a second exemplary view for describing various types of coupling between the transporting unit and the cooling plate constituting the substrate transporting device according to an embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(12) Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings. Advantages and features of the present invention, and methods for achieving them will be clarified with reference to embodiments described below in detail together with the accompanying drawings. However, the present invention is not limited to the embodiments disclosed below, but may be implemented in various different forms, and only the embodiments allow the publication of the present invention to be complete, and are provided to fully inform those skilled in the technical field to which the present invention pertains of the scope of the invention, and the invention is only defined by the scope of the claims. The same reference numerals refer to the same components throughout the specification.

(13) When elements or layers are referred to as “on” or “above” of other elements or layers, it includes not only when directly above of the other elements or layers, but also other layer or other element intervened in the middle. On the other hand, when elements are referred to as “directly on” or “directly above,” it indicates that no other element or layer is intervened therebetween.

(14) The spatially relative terms “below,” “beneath,” “lower,” “above,” “upper,” etc., as shown in figures, can be used to easily describe the correlation of an element or components with other elements or components. The spatially relative terms should be understood as terms including the different direction of the element in use or operation in addition to the direction shown in the drawing. For example, if the element shown in the figure is turned over, an element described as “below” or “beneath” the other element may be placed “above” the other element. Accordingly, the exemplary term “below” can include both the directions of below and above. The element can also be oriented in other directions, so that spatially relative terms can be interpreted according to the orientation.

(15) Although the first, second, etc. are used to describe various elements, components and/or sections, these elements, components and/or sections are not limited by these terms. These terms are only used to distinguish one element, component or section from another element, component or section. Therefore, the first element, first component or first section mentioned below may be a second element, second component or second section within the technical spirit of the present invention.

(16) The terminology used herein is for describing the embodiments and is not intended to limit the present invention. In the present specification, the singular form also includes the plural form unless otherwise specified in the phrase. As used herein, “comprises” and/or “comprising” means that the components, steps, operations and/or elements mentioned above do not exclude the presence or additions of one or more other components, steps, operations and/or elements.

(17) Unless otherwise defined, all terms (including technical and scientific terms) used in the present specification may be used as meanings commonly understood by those skilled in the art, to which the present invention pertains. In addition, terms defined in the commonly used dictionary are not ideally or excessively interpreted unless specifically defined.

(18) Hereinafter, exemplary embodiments of the present invention will be described in detail with reference to the accompanying drawings, and in the description with reference to the accompanying drawings, the same or corresponding components are assigned the same reference numbers regardless of reference numerals, and the description overlapped therewith will be omitted.

(19) When the temperature of a transporting robot that transports a substrate between a plurality of process chambers increases, the driving performance of the transporting robot may be deteriorated, and the lifespan of the transporting robot may be shortened.

(20) Although the transporting robot may be cooled using airflow, since the substrate treating

system including a plurality of process chambers and the transporting robot is operated in a vacuum state, a cooling method using airflow is difficult to apply.

(21) In the present invention, a substrate treating system capable of preventing an increase in the temperature of the transporting robot (e.g., the surface temperature of the transporting robot, the internal temperature of the transporting robot, etc.) by installing a cooling plate around the transporting robot and a substrate transporting device included in such system are provided.

(22) Hereinafter, the present invention will be described in detail with reference to the drawings.

(23) FIG. 1 is a plan view schematically showing the internal structure of a substrate treating system including a substrate transporting device.

(24) Referring to FIG. 1, a substrate treating system **100** may include an index module **110** and a process processing module **120**.

(25) The index module **110** and the process processing module **120** may be sequentially arranged in one direction. In the present embodiment, the direction, in which the index module **110** and the process processing module **120** are arranged, is defined as the first direction **10**. Further, a direction perpendicular to the first direction **10** when viewed from above is defined as the second direction **20**, and a direction perpendicular to the plane including the first direction **10** and the second direction **20** is defined as the third direction **30**.

(26) The index module **110** is arranged in front of the process processing module **120**. The index module **110** may include a load port **111** and a transporting frame **112**.

(27) A carrier **130**, in which the substrate is accommodated, is mounted on the load port **111**. A plurality of such load ports **111** may be provided in front of the transporting frame **112**, and a plurality of load ports **111** may be arranged in a row along the second direction **20**.

(28) In FIG. 1, it is shown that the index module **110** is provided with four load ports **111**. However, this embodiment is not limited thereto. The number of load ports **111** may increase or decrease according to conditions such as process efficiency of the process processing module **120** and foot-print.

(29) The carrier **130** accommodates a plurality of substrates (e.g., wafers). The carrier **130** may have a slot (not shown) provided therein to support the edge of the substrate.

(30) A plurality of slots may be provided in the third direction **30**. In this case, the substrates may be stacked in a state spaced apart from each other along the third direction **30** and positioned in the carrier **130**. The carrier **130** may be implemented as, for example, a front opening unified pod (FOUP).

(31) The transporting frame **112** transports the substrate between the carrier **130** mounted on the load port **111** and the buffer unit **121**. The transporting frame **112** may include an index rail **113** and an index robot **114**.

(32) The index rail **113** provides a path through which the index robot **114** moves. The index rail **113** may be arranged such that its longitudinal direction is parallel to the second direction **20**.

(33) The index robot **114** directly transports the substrate. The index robot **114** may be installed on the index rail **113** and may linearly move along the second direction **20** on the index rail **113**.

(34) The index robot **114** may include a first base **114a**, a first body **114b**, and an index arm **114c**. Here, the first base **114a** may be installed to be movable along the index rail **113**.

(35) The first body **114b** may be coupled to the first base **114a**. The first body **114b** may be provided to be movable along the third direction **30** on the first base **114a**. Further, the first body **114b** may be provided to be rotatable on the first base **114a**.

(36) The index arm **114c** is coupled to the first body **114b**, and may be provided to be movable forward and backward with respect to the first body **114b**. A plurality of such index arms **114c** may be provided on the first body **114b** to be individually driven.

(37) The plurality of index arms **114c** may be stacked and arranged to be spaced apart from each other along the third direction **30**. Some of the plurality of index arms **114c** may be used when transporting the substrate from the process processing module **120** to the carrier **130**, and other

parts may be used when transporting the substrate from the carrier **130** to the process processing module **120**. When the plurality of index arms **114c** are configured in this way, it is possible to obtain an effect of preventing particles generated from the substrate before process treating from adhering to the substrate after process treating in the process of the index robot **114** carrying in and out the substrate.

(38) The process processing module **120** may include a buffer unit **121**, a substrate transporting device **122**, and a process chamber **123**.

(39) The buffer unit **121** provides a space, in which the substrate stays between the transporting frame **112** and the substrate transporting device **122** before the substrate is transported. The buffer unit **121** may be arranged between the transporting frame **112** and the substrate transporting device **122** for this purpose.

(40) The buffer unit **121** may include a slot (not shown) therein, on which the substrate is mounted. A plurality of slots may be provided in the buffer unit **121**, and a plurality of slots may be provided to be spaced apart from each other along the third direction **30**. On the other hand, in the buffer unit **121**, the surface facing the transporting frame **112**, the surface facing the substrate transporting device **122**, and the like may be opened respectively.

(41) The substrate transporting device **122** transports a substrate between the buffer unit **121** and the process chamber **123**. The substrate transporting device **122** may include a guide rail and a transporting unit. A more detailed description of the substrate transporting device **122** will be described later with reference to FIG. 2.

(42) Meanwhile, the substrate transporting device **122** may transport a substrate between two different process chambers **123**.

(43) The substrate transporting device **122** may be arranged in that its longitudinal direction is parallel to the first direction **10**. At this time, a plurality of process chambers **123** may be arranged on both sides of the substrate transporting device **122** along the second direction **20**, and a plurality of process chambers **123** may be arranged on each side of the substrate transporting device **122** along the first direction **10**.

(44) The plurality of process chambers **123** may be arranged to be stacked with each other. That is, the plurality of process chambers **123** may be arranged in an X*Y arrangement on one side of the substrate transporting device **122**. Here, X is a natural number of 1 or more, which means the number of process chambers **123** provided in a row along the first direction **10**, and Y is a natural number of 1 or more, which means the number of process chambers **123** provided in a row along the third direction **30**.

(45) For example, when four process chambers **123** are provided on one side of the substrate transporting device **122**, the four process chambers **123** may be arranged in 2*2 arrangement, and when six process chambers **123** are provided on one side of the substrate transporting device **122**, the six process chambers **123** may be arranged in 3*2 arrangement.

(46) Meanwhile, the number of process chambers **123** may increase or decrease. Further, the process chamber **123** may be provided only on one side of the substrate transporting device **122**, or may be provided in a single layer on one or both sides of the substrate transporting device **122**.

(47) The process chamber **123** may be implemented as an etching chamber for etching a substrate in this embodiment. However, this embodiment is not limited thereto. The process chamber **123** may be implemented as a cleaning chamber for cleaning the substrate.

(48) Next, the substrate transporting device **122** will be described. FIG. 2 is a perspective view showing a detailed structure of the substrate transporting device shown in FIG. 1.

(49) According to FIG. 2, the substrate transporting device **122** may include a transporting unit **210**, a guide rail **220**, and a support block **230**.

(50) The substrate transporting device **122** transports a substrate W (e.g., wafer). The substrate transporting device **122** may transport a substrate between the buffer unit **121** and the process chamber **123**, or between a first chamber, which is one of a plurality of process chambers **123** and a

second chamber, which is another one.

(51) Both the first chamber and the second chamber may be implemented as an etching chamber. However, this embodiment is not limited thereto. One of the first chamber and the second chamber may be implemented as an etching chamber, and the other chamber may be implemented as a cleaning chamber.

(52) The transporting unit **210** performs a linear motion between the buffer unit **121** and the process chamber **123** or between the first chamber and the second chamber to transport the substrate W. This transporting unit **210** may be implemented as a transporting robot.

(53) The transporting unit **210** may be installed on the guide rail **220**. The transporting unit **210** may perform linear motion along the first direction **10** on the guide rail **220** in a state, in which the substrate W is loaded.

(54) The transporting unit **210** may include a second base **211**, a second body **212**, a driving member **213** and an arm member **214**.

(55) The second base **211** has a flat plate shape and may be installed in the lower portion of the transporting unit **210**. The second base **211** may be installed to be movable in the first direction **10** along the guide rail **220**.

(56) The second base **211** may be installed on the support block **230**. In this case, the second base **211** may have a tube therein to electrically connect the support block **230** and the motor.

(57) The path provided inside the second base **211** may be formed of an electrically conductive material (e.g., a metal such as copper) as a material. However, this embodiment is not limited thereto. The second base **211** may have a transmission line installed therein to electrically connect the support block **230** and the motor.

(58) The second base **211** may be installed on a plurality of support blocks **230** arranged at predetermined intervals. For example, the second base **211** may be installed on four support blocks **230** arranged at a predetermined interval in the first direction **10** and the second direction **20**.

(59) The second body **212** constitutes the body of the transporting unit **210** and may be installed by being coupled to the second base **211**. The second body **212** may be installed to support the driving member **213** and the arm member **214**.

(60) The driving member **213** moves the arm member **214** to a predetermined position. The driving member **213** may be provided on the second body **212** so as to be rotatable in the second direction **20**. Further, the driving member **213** may be provided on the second body **212** so as to be able to elevate in the third direction **30**.

(61) The arm member **214** loads the substrate W, and may unload the substrate W to a predetermined position under the control of the driving member **213**.

(62) A single arm member **214** may be installed on the driving member **213**. In this case, the arm member **214** may be installed by being coupled to the upper plate of the driving member **213** in the lateral direction. However, this embodiment is not limited thereto. A plurality of arm members **214** may be installed on the driving member **213**.

(63) When a plurality of arm members **214** are installed on the driving member **213**, the arm members **214** may be arranged in a vertical direction (i.e., the third direction **30**) in a state spaced apart from each other on the side surface of the driving member **213**. When a plurality of arm members **214** are installed on the driving member **213**, it is also possible for the arm members **214** to be arranged in the left and right directions (i.e., the first direction **10** or the second direction **20**) on the side surface of the driving member **213**.

(64) When a plurality of arm members **214** are installed on the driving member **213**, some of the plurality of arm members **214** may be used when transporting a substrate from the buffer unit **121** to the process chamber **123**, and other parts may be used when transporting the substrate from the process chamber **123** to the buffer unit **121**.

(65) The guide rail **220** is to provide a path, through which the transporting unit **210** moves. The guide rail **220** may be provided in that its longitudinal direction is the first direction **10**, and may be

installed as a pair of rails **221** and **222**, that is, the first rail **221** and the second rail **222**.

(66) The support block **230** is coupled to the upper portion of the guide rail **220**. The support block **230** may be installed so as to be slidable in the first direction **10** along the guide rail **220**. The transporting unit **210** may be installed on the support block **230**.

(67) A plurality of support blocks **230** may be installed on the guide rail **220**. In this case, a plurality of support blocks **230** may be installed on each of the first rail **221** and the second rail **222** constituting the guide rail **220**.

(68) When a plurality of support blocks **230** are installed on each of the first rail **221** and the second rail **222**, the same number may be installed on each of the first rail **221** and the second rail **222**. For example, two support blocks **230** may be installed on each of the first rail **221** and the second rail **222**.

(69) The support block **230** may include a first block **231**, a second block **232**, a third block **233**, and a fourth block (not shown). The first block **231** and the second block **232** are installed on the first rail **221** and may be installed spaced apart in the first direction **10**. The third block **233** and the fourth block are installed on the second rail **222** and may be installed spaced apart in the first direction **10** as in the case of the first block **231** and the second block **232**.

(70) The third block **233** may be formed at the position corresponding to the first block **231**. That is, the third block **233** may be formed at the position parallel to the first block **231** in the second direction **20**. Although not shown in FIG. 2, the second block **232** and the fourth block may also be formed in the same shape as the first block **231** and the third block **233**.

(71) In the above description, the substrate transporting device **122** is defined as transporting a substrate between the buffer unit **121** and the process chamber **123** (or between the first chamber and the second chamber). However, this embodiment is not limited thereto. The transporting frame **112** for transporting the substrate between the carrier **130** mounted on the load port **111** and the buffer unit **121** may also be implemented as a substrate transporting device.

(72) The substrate transporting device **122** may include a cooling plate around the transporting unit **210** to control the temperature of the transporting unit **210**. This will be described below.

(73) FIG. 3 is a plan view schematically showing a substrate transporting device including a cooling plate according to an embodiment of the present invention, and FIG. 4 is a cross-sectional view schematically showing a substrate transporting device including a cooling plate according to an embodiment of the present invention.

(74) According to FIGS. 3 and 4, the substrate transporting device **122** may be configured to include a transporting unit **210**, a guide rail **220**, a cooling plate **310**, a pump **320**, a cooling flow path **330**, and a valve **340**.

(75) Since the transporting unit **210** and the guide rail **220** have been described above with reference to FIG. 2, detailed descriptions thereof will be omitted herein. Further, although not shown in FIGS. 3 and 4, the substrate transporting device **122** may be configured to include a support block **230**.

(76) The cooling plate **310** is for controlling the temperature of the transporting unit **210** (e.g., the surface temperature of the transporting unit **210**, the internal temperature of the transporting unit **210**, etc.). The cooling plate **310** has a cooling flow path **330** through which coolant can flow, and the temperature of the transporting unit **210** can be controlled through the coolant passing through the cooling flow path **330**.

(77) As shown in FIG. 5, the cooling plate **310** may include a first plate member **410**, a pipe member **420**, and a second plate member **430**. FIG. 5 is a perspective view showing a detailed structure of a cooling plate constituting a substrate transporting device according to an embodiment of the present invention.

(78) The body constitutes the body of the cooling plate **310**. Such body may be formed by combining the first plate member **410** and the second plate member **430**.

(79) The first plate member **410** is arranged on one side surface of the pipe member **420**. The first

plate member **410** is formed to cover one side surface of the pipe member **420**, and may be formed of an appropriate material and installed so as to discharge cold air emitted from the pipe member **420** to the outside.

(80) The second plate member **430** is arranged on the other side surface of the pipe member **420**. The second plate member **430** may be formed to cover the other side surface of the pipe member **420**, and may be formed to seal the pipe member **420** by being combined with the first plate member **410**. Like the first plate member **410**, the second plate member **430** may be formed of an appropriate material and installed so as to discharge cool air emitted from the pipe member **420** to the outside.

(81) The pipe member **420** provides a flow path through which coolant can flow. The pipe member **420** is arranged inside the cooling plate **310** and may be installed so that its outer shape is not exposed to the outside by the first plate member **410** and the second plate member **430**.

(82) The pipe member **420** may be installed in a zigzag shape from the top to the bottom direction (i.e., the negative third direction **30**) as shown in FIG. **6** inside the cooling plate **310**. However, this embodiment is not limited thereto. The pipe member **420** may be installed in a zigzag shape from one side to the other side direction (i.e., the first direction **10**) as shown in FIG. **7**. FIG. **6** is a first exemplary view showing various installation types of the pipe members constituting the cooling plate shown in FIG. **5**, and FIG. **7** is a second exemplary view showing various installation types of the pipe members constituting the cooling plate shown in FIG. **5**.

(83) On the other hand, the cooling plate **310** does not have a separate pipe member **420**, and may form a groove that can be used as a flow path through which coolant flows in at least one member of the first plate member **410** and the second plate member **430**.

(84) Meanwhile, in this embodiment, in order to increase the cooling efficiency of the cooling plate **310**, a thermal compound may be applied to the surface of the pipe member **420** or a thermal pad may be mounted on the surface of the pipe member **420**. Further, in the present embodiment, it is possible to apply thermal grease to the surface of the pipe member **420**.

(85) It will be described again with reference to FIG. **3**.

(86) The cooling plate **310** may be installed on the side surface of the transporting unit **210**. The cooling plate **310** may be installed on both side surfaces of the transporting unit **210**, and may be installed only on one side surface of the transporting unit **210**.

(87) When the cooling plate **310** is installed on the side surface of the transporting unit **210** in this way, although the high temperature heat energy **510** is discharged from the plurality of process chambers **123** arranged on both sides of the transporting unit **210** as shown in FIG. **8**, it is possible to prevent the temperature of the transporting unit **210** from being increased by the cooling plate **310**. FIG. **8** is a reference diagram for describing the performance of a substrate transporting device having a cooling plate according to an embodiment of the present invention.

(88) Meanwhile, the cooling plate **310** may be installed not only at the side surface of the transporting unit **210** but also in the lower portion of the transporting unit **210** as shown in FIG. **9**. FIG. **9** is a cross-sectional view schematically illustrating a substrate transporting device including a cooling plate according to another embodiment of the present invention.

(89) It will be described again with reference to FIG. **3**.

(90) The pump **320** supplies coolant to the cooling flow path **330**. The pump **320** may be installed outside the cooling plate **310** to supply coolant to the cooling flow path **330** (i.e., the pipe member **420**) formed to extend into the cooling plate **310**.

(91) The cooling flow path **330** provides a path through which coolant may flow. This cooling flow path **330** is connected to the pump **320**, and may be formed passing through the inside of the cooling plate **310** through an inlet portion **331** and an outlet portion **332** positioned inside the cooling plate **310**. The coolant supplied by the pump **320** may be discharged to the outside by passing through the cooling flow path **330** installed inside the cooling plate **310** in the form of a pipe member **420**. In this embodiment, PCW (Process Cooling Water) may be used as coolant.

(92) The valve **340** controls the flow of coolant. The valve **340** is installed on the cooling flow path **330** to open and close the cooling flow path **330**, thereby controlling the flow of coolant.

(93) Meanwhile, in this embodiment, the cooling plate **310** is not limited to being installed spaced apart from the transporting unit **210** at a predetermined distance as shown in FIG. **3**. For example, it is also possible for the cooling plate **310** to be installed being attached to the side surface of the transporting unit **210**. When the cooling plate **310** is installed in close contact with the side surface of the transporting unit **210** as described above, an effect of improving the cooling efficiency by the cooling plate **310** can be obtained.

(94) When the cooling plate **310** is formed in close contact with the side surface of the transporting unit **210**, the cooling plate **310** may be coupled to the side surface of the transporting unit **210** by using the first protrusion **610** formed on the side surface of the transporting unit **210** and the second protrusion **620** formed on the side surface of the cooling plate **310**, as shown in FIG. **10**.

(95) In this case, the second protrusion **620** may be formed on the side surface of the cooling plate **310** facing the side surface of the transporting unit **210**, and may be formed at different heights to cross the first protrusion **610**. FIG. **10** is a first exemplary view for describing various types of coupling between a transporting unit and a cooling plate constituting a substrate transporting device according to an embodiment of the present invention.

(96) However, this embodiment is not limited thereto. When the cooling plate **310** is formed in close contact with the side surface of the transporting unit **210**, it is also possible to couple the cooling plate **310** to the side surface of the transporting unit **210** using the third protrusion **630** formed on the side surface of the transporting unit **201** and the groove **640** formed on the side surface and the cooling plate **310** as shown in FIG. **11**. In this case, the third protrusion **630** may be inserted into the groove **640** to couple the cooling plate **310** to the side surface of the transporting unit **210**. FIG. **11** is a second exemplary view for describing various types of coupling between the transporting unit and the cooling plate constituting the substrate transporting device according to an embodiment of the present invention.

(97) Meanwhile, although not shown, in this embodiment, the cooling plate **310** may be coupled to the side surface of the transporting unit **210** using a groove formed on the side surface of the transporting unit **210** and a fourth protrusion formed on the side surface of the cooling plate **310**, and it is also possible to couple the cooling plate **310** to the side surface of the transporting unit through bolting, welding, or the like.

(98) The substrate transporting device **122** according to various embodiments of the present invention has been described above with reference to FIGS. **3** to **11**. The present invention relates to a method of improving the thermal efficiency of a cooling plate of Liner TM. The cooling plate **310** is mounted on the left/right side surface or the lower portion of the transporting unit **210**, and a cooling effect due to heat generation of the transporting unit **210** using the phenomenon of thermal radiation can be obtained.

(99) The cooling plate **310** may be mounted in the lower portion of the traveling shaft of the transporting unit **210**, on the left/right side surface, or in both the lower portion/left side surface/right side surface depending on the situation, and the cooling plate **310** can function to suppress the increase of TM temperature by the process chamber **123**.

(100) The cooling plate **310** may be configured in a structure, in which coolant can flow therein. The cooling plate **310** may have a structure, in which coolant can flow using a separate pipe, or form a coolant path through a groove inside the plate.

(101) Thermal efficiency is very important when coolant is injected into the cooling plate **310** using a pipe. To this end, a thermal compound may be applied to a cooling pipe formed inside the cooling plate **310** to improve thermal efficiency between the plate and the pipe. Alternatively, thermal efficiency can be improved by mounting a thermal PAD on the cooling pipe. Alternatively, thermal efficiency can be improved by applying thermal grease to the cooling pipe.

(102) The present invention also proposes a method of mounting between the transporting unit **210**

and the cooling plate **310** for efficient heat exchange.

(103) The cooling plate **310** and the robot body of the transporting unit **210** may have a structure, in which the blades cross each other as shown in FIG. **10**, in order to increase the heat exchange effect. The heat exchange effect can be maximized by maximizing the crossing area between the cooling plate **310** and the robot body of the transporting unit **210** through the wing structure.

(104) The coolant in the cooling plate **310** may have a structure that is circulated as shown in FIGS. **6** and **7**. PCW supplied from a semiconductor line may be used as coolant.

(105) Although the embodiments of the present invention have been described with reference to the above and the accompanying drawings, those of ordinary skill in the art to which the present invention pertains can understand that the present invention can be implemented in other specific forms without changing the technical spirit or essential features. Therefore, it should be understood that the embodiments described above are illustrative and non-limiting in all respects.

Claims

1. A system for treating a substrate comprising: a substrate transporting apparatus operated in a vacuum state; and a plurality of process chambers disposed along a first direction and disposed on both sides of the substrate transporting apparatus along a second direction perpendicular to the first direction, wherein the substrate transporting apparatus comprises: a transporting robot loading a substrate into the plurality of process chambers or unloading the substrate from the plurality of process chambers, and linearly moving along the first direction; a guide rail provided as a moving path of the transporting robot and disposed parallel to the first direction; and a cooling plate installed as a side wall extending along the guide rail and a flow path, through which coolant flows, is installed inside the cooling plate.
2. The system for treating a substrate of claim 1, wherein the transporting robot comprises: a base installed to be movable along the guide rail; a driving member installed on the base and provided on the base so as to be able to elevate along a third direction perpendicular to a plane including the first direction and the second direction; and an arm member installed on the drive member and loading or unloading the substrate, and provided rotatably.
3. The system for treating a substrate of claim 2, wherein the cooling plate is installed as the side wall along the third direction in addition to the first direction to control the temperature of the transporting robot.
4. The system for treating a substrate of claim 3, wherein the guide rail includes a first rail and a second rail, and the cooling plate disposed in the substrate transporting apparatus so as to be adjacent to both the first rail and the second rail along the first direction.
5. The system for treating a substrate of claim 4, wherein the cooling plate comprises: a pipe member provided as the flow path, through which coolant flows; and a first plate member and a second plate member coupled to each other on both sides of the pipe member so that the pipe member can be embedded therein, and the first plate member, the second plate member, and the pipe member are formed separably.
6. The system for treating a substrate of claim 4, wherein the cooling plate comprises: the flow path is provided as a groove formed on at least one surface of a first plate, a second plate, a first surface of a first plate member and a second surface of a second plate member facing the first surface.
7. The system for treating a substrate of claim 4, wherein the flow path is installed in a zigzag pattern inside the cooling plate.
8. The system for treating a substrate of claim 4, wherein at least one of a thermal compound, a thermal pad, and a thermal grease is installed on the flow path.
9. The system for treating a substrate of claim 4, wherein the substrate transporting apparatus comprises: a pump for supplying coolant; a cooling flow path provided as a passage through which the coolant flows, and connected to the flow path installed inside the pump and the cooling plate;

and a valve installed on the cooling flow path and for opening and closing the cooling flow path.
10. The system for treating a substrate of claim 4, wherein the coolant is PCW.
